

# Daniel (Chenyu) Lin

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## EDUCATION

### Cornell University

Ithaca, NY

*B.A. in Computer Science and Mathematics (double major)*

Aug 2024 – Expected May 2027

### University of North Carolina at Chapel Hill

Chapel Hill, NC

*Transferred Out, Computer Science*

Aug 2023 – May 2024

### Selected Courseworks:

AI/ML: Machine Learning | Large-Scale ML | Natural Language Processing | Reinforcement Learning (ongoing) | Computational Optimization (grad, ongoing)

Maths: Information Theory (grad) | Measure Theory (grad, ongoing) | Stochastic Process | (Advanced) Linear Algebra | Honors Analysis

General CS: Algorithms | Functional Programming | Data Structures | Embedded System

## RESEARCH INTEREST

My research centers on **unsupervised and weakly supervised learning**, with **music and audio technologies** as my primary downstream domain. I am drawn to systems that discover structure on their own, and to what such systems can offer creative work that more heavily supervised approaches cannot. I am also interested in the theory underlying these methods, and in the mathematical structure that determines what unsupervised learning can recover.

## PUBLICATIONS

- Daniel Chenyu Lin, Michael Freeman, John Thickstun. “Assessing Factual Music Comprehension in Large Audio Language Models.” *International Society for Music Information Retrieval (ISMIR)*, 2026.
- Saebyeol Shin, Chao Wan, Zhenzhen Liu, Justin Lovelace, Daniel C. Lin, Kilian Q. Weinberger, John Thickstun. “Music Transcription with (Almost) No Supervision.” *In submission*.

## RESEARCH EXPERIENCE

### Cornell Bowers CIS | Advisor: Dr. John Thickstun

Ithaca, NY

*Student Researcher, Generative Models & Music Information Retrieval*

Oct 2024 – Present

*Oct 2024 – Apr 2026*

- **Factual evaluation of large audio language models**  
Showed that standard text-similarity evaluation on MusicQA cannot distinguish LALM responses conditioned on correct audio from responses conditioned on random audio. Designed a factual evaluation protocol that parses open-ended responses into structured labels scored by precision/recall/F1, and benchmarked nine frontier and open-source models across six factual retrieval tasks on MusicNet, FMA, and a newly curated dataset. Awarded a \$6,000 research grant from Cornell’s College of Arts and Sciences. (First author; accepted to ISMIR 2026)
- **Unsupervised automatic music transcription** | Co-advisor: Dr. Kilian Q. Weinberger *Oct 2025 – May 2026*  
Co-authored a study of unsupervised transcription with light supervision to break the symmetry on symbolic marginals; built the training corpus, scraping and normalizing ten sources into ~440 hours of audio and ~57 hours of symbolic data across four annotation formats; sole undergraduate on a team of four PhD students. (In submission)
- **Symmetry and identifiability in unsupervised translation** *May 2026 – Aug 2026*  
Led the theoretical analysis of why unsupervised transcription recovers the target map only up to a constant pitch shift, characterizing the symmetry group of the symbolic marginal and deriving deterministic and variational cycle-consistent objectives with gradient penalties for a controlled 2D translation testbed, then analyzing why these objectives fail to isolate the intended solution.
- **Instruction tuning LLMs for MIDI understanding** | Co-advisor: Dr. Chris Donahue *Aug 2026 – Present*  
Leading the instruction-tuning and reinforcement-learning stages of a symbolic-music language model, working alongside two PhD students, with a focus on what musical understanding emerges without task-specific supervision.

## Lyra Lab, Tencent Music Entertainment

Shenzhen, China

Research Intern, Audio Information Retrieval

Jun 2024 – Aug 2024

- Replaced the Conformer encoder in a production Cover Song Identification system with a Zipformer encoder, raising top-1 accuracy by 7 points to over 90%, and resolved exploding gradients at scale by reimplementing the block's activation function.
- Pre-trained MT3 from scratch on roughly twice the data used in the original paper, improving instrument detection and onset alignment.
- Built an in-batch audio augmentation pipeline supporting speed and pitch variation, noise injection, and impulse response convolution.

Music Tech X (High School Program) Student Researcher & Group Leader

Jun 2023 – Aug 2023

- Implemented an audio fingerprinting algorithm from scratch for music recognition, and led a team of six in improving its search and evaluation components to reach 96% precision on the MIREX dataset.
- Explored a learned alternative to fingerprinting, training a ResNet to recognize pitch-shifted and tempo-varied music with over 80% precision.

## TEACHING & ACADEMIC INVOLVEMENT

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### Reading Group Participant & Presenter

Cornell University & Ithaca College

Ithaca, NY

- Theory of Generative Modeling (CS 7792)
- Music Technology

Spring 2026

Fall 2024 – Spring 2025

### Teaching Assistant

Cornell Bowers CIS

Ithaca, NY

- ECE 5620: Information Theory and Generative Modeling
- CS 4782: Introduction to Deep Learning
- CS 3780: Introduction to Machine Learning
- CS 1109: Foundational Programming Concepts

Fall 2026

Spring 2026

Fall 2025

Summer 2025

## TECHNICAL SKILLS

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**Programming Languages:** Python, Java, OCaml, C (cuBLAS), ARM Assembly

**Machine Learning:** Pytorch, Tensorflow, Huggingface, W&B, Data Augmentations

**Audio Programming:** Librosa, Torchaudio, Madmom, Muscore, Logic, Audacity

**Other Useful Skills:** Linux, Slurm Cluster, GPU Deployment, Github, Google Colab